18. Slide Verification Problem #18

Slope, homogenous slope, ru pore pressure

18.1. Introduction

This model is taken from Baker (1980) and was originally published by Spencer (1969). It consists of a simple slope of homogeneous soil with pore pressure.

18.2. Problem Description

Verification problem #18 is shown in Figure 18.1. The material properties are given in Table 18.1. The position of the critical slip surface and the corresponding factor of safety is calculated for a noncircular slip surface. The pore pressure within the slope is modeled using a Ru value of 0.5.

18.3. Geometry and Material Properties

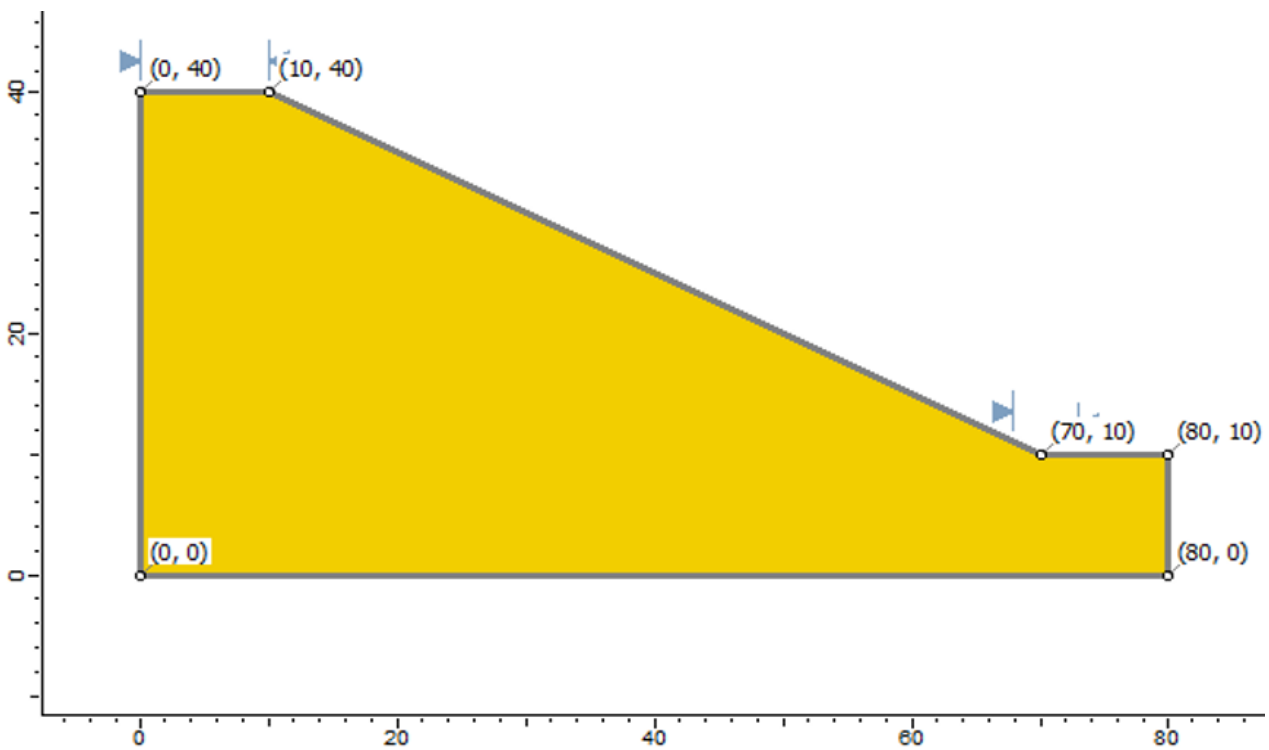

Figure 18.1: Geometry Setup in **Slide**

Table 18.1: Soil Properties

	c´ (kN/m^2)	ϕ´ (deg.)	γ (kN/m^3)	Ru
Soil	10.8	40	18	0.5

18.4. Results

Table 18.2: Noncircular – using Random search with Monte-Carlo optimization

Method	Factor of Safety
Spencer	1.010

Note: Baker (1980) Spencer factor of safety = 1.02

Spencer (1969) Spencer factor of safety = 1.08

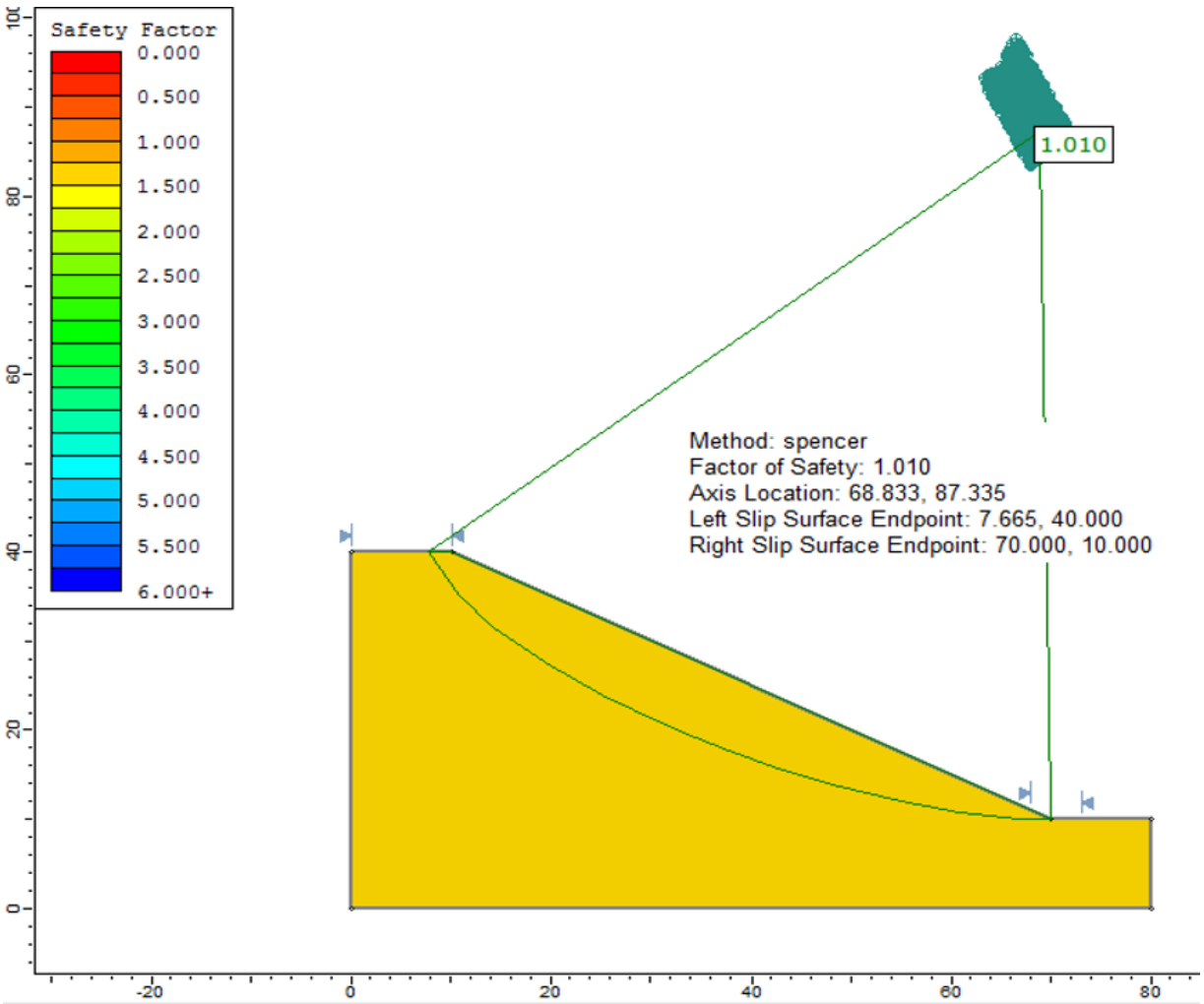

Figure 18.2: Noncircular failure surface, using the Spencer method